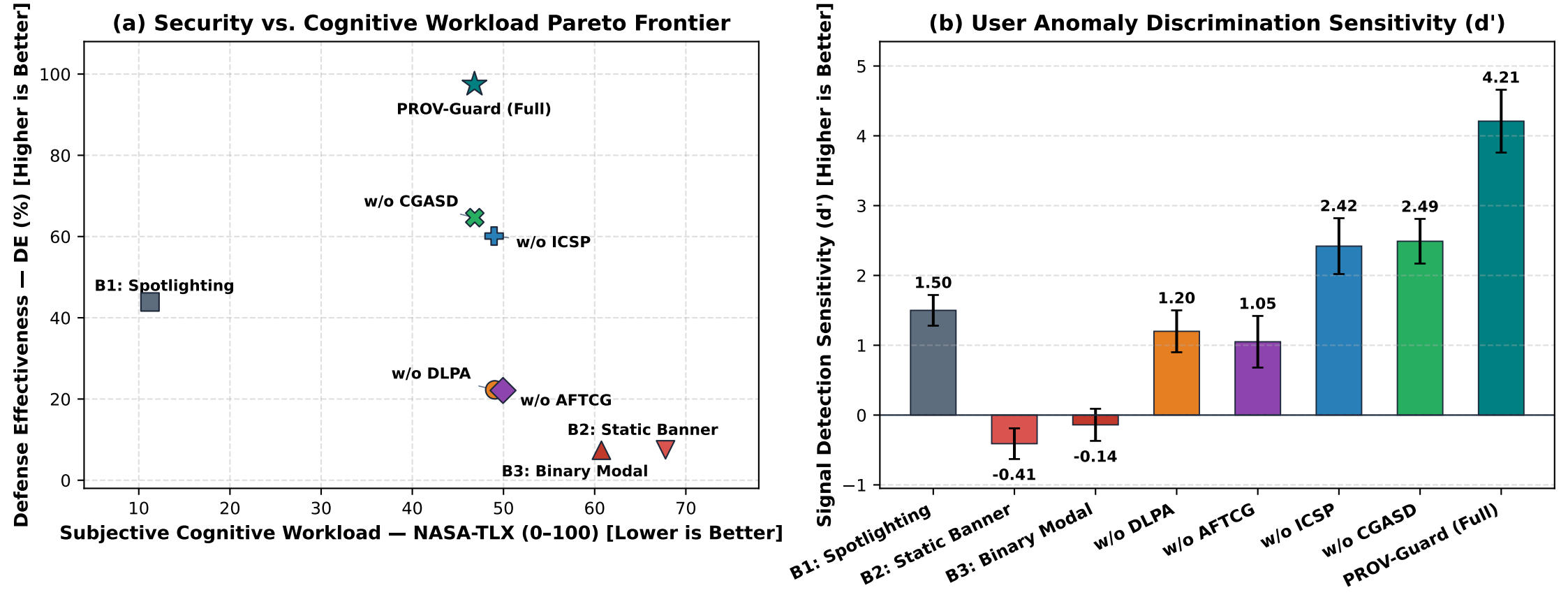
(a) Security vs. Cognitive Workload Pareto Frontier
Defense Effectiveness — DE (%) [Higher is Better]
Subjective Cognitive Workload — NASA-TLX (0–100) [Lower is Better]
PROV-Guard (Full)
w/o CGASD
w/o ICSP
B1: Spotlighting
w/o DLPA
w/o AFTCG
B2: Static Banner
B3: Binary Modal
(b) User Anomaly Discrimination Sensitivity (d')
Signal Detection Sensitivity (d') [Higher is Better]
1.50
-0.41
-0.14
1.20
1.05
2.42
2.49
4.21
B1: Spotlighting
B2: Static Banner
B3: Binary Modal
w/o DLPA
w/o AFTCG
w/o ICSP
w/o CGASD
PROV-Guard (Full)